

Academic Org: Div of Material Science & Erg – Subject: Advanced Materials Science

Course: AMSC5738 **Course ID:** 015565 **Eff Date:** 2026-07-01 **Crse Status:** Active **Apprv. Status:** Pending **[New Course]**
Surface Science and Thin Film Technology 表面科學和薄膜技術

This course introduces surface science, the fabrication, characterization, and physical properties of thin films. The topics include vacuum science and technology, basic concepts of surface science, thin film deposition techniques, kinetics and thermodynamics of surface reactions, film growth processes and modes, characterization, epitaxy, and physical properties of thin films. State-of-the-art research advancements on thin film preparation and properties are also selectively introduced.

本科主要介紹表面科學、薄膜加工製備表徵方法和相關的物理性質，內容包括：真空知識和技術、表面科學基礎知識、各種鍍膜技術、薄膜生長過程機理、薄膜基本特性的測量、外延生長、半導體複雜結構合成工程、以及薄膜的光學和電學性質。本科也將有選擇地介紹當前表面科學和薄膜科研的前沿進展。

Grade Descriptor: A

Demonstrate thorough mastery of principles and subject matter in the course required for attaining all the course learning outcomes. Demonstrate the ability to apply the principles or subject matter to familiar and unfamiliar situations, in a manner that would surpass the normal expectation at this level and standards that may be required at higher levels of study or research. Has the ability to express the knowledge or synthesis of ideas in a clear and cogent manner.

完全掌握科目內容中的原理及知識以達到所有學習目標。能夠將科目中的原理或知識，以超出在該程度的一般要求及在更高水平的課程或研究可能需要的標準，應用於熟悉和陌生的情況。有能力以清晰及具說服力的方式表達科目知識或綜合概念。

A-

Demonstrate substantial command of principles and subject matter in the course required for attaining almost all the course learning outcomes. Demonstrate the ability to state and apply the principles or subject matter to familiar and some unfamiliar situations, in a manner that is logical and comprehensive. Has the ability to express the knowledge or application with clarity and accuracy.

充分掌握科目內容中的原理及知識以達到絕大部份學習目標。能夠將科目中的原理或知識，以合乎邏輯和全面的方式，應用於熟悉和一些陌生的情況。有能力清晰和準確地表達科目知識或應用。

B

Demonstrate general and sufficient command of principles and subject matter in the course required for attaining most of the course learning outcomes. Demonstrate the ability to state and apply the principles or subject matter accurately to most (but not necessarily all) familiar and standard situations, in a manner that is logical and persuasive. Has the ability to express the knowledge or application in a satisfactory and unambiguous way.

一般並足夠地掌握科目內容中的原理及知識以達到大部份學習目標。能夠將科目中的原理或知識，以合乎邏輯和有說服力的方式，精確地陳述及應用於大部份（但不一定是所有）熟悉和標準的情況。有能力以令人滿意和明確的方式表達科目知識或應用。

C

Demonstrate general command of principles and subject matter in the course required for attaining some of the course learning outcomes. Demonstrate the ability to state and apply the principles or subject matter to most (but not necessarily all) familiar and standard situations, but with occasional errors and/or in a manner that is fragmented. Has the ability to express the separate pieces of knowledge in an unambiguous way.

一般地掌握科目內容中的原理及知識以達到部份學習目標。雖然偶有錯誤，但能夠在大部份（但不一定是所有）熟悉和標準的情況下，零碎地陳述和應用科目中的原理或知識。能明確地表達個別部分的科目知識。

D

Demonstrate partial command of principles and subject matter in the course required for attaining some of the course learning outcomes. Demonstrate the ability to state and apply the principles or subject matter to some simple situations only. Has the ability to state the knowledge or application in simple terms.

部份地掌握科目內容中的原理及知識以達到部份學習目標。只能夠在一些簡單情況陳述和應用科目中的原理或知識。能以簡單的用語陳述科目知識或應用。

F

Demonstrate little or no evidence of command of principles and subject matter in the course required for attaining the course learning outcomes, OR failure to meet specified assessment requirements.

很少、甚至沒有掌握科目內容中的原理及知識以達到學習目標，或者沒有達到規定的評估要求。

Equivalent Offering:

Units: 3 (Min) / 3 (Max) / 3 (Acad Progress)

Grading Basis: Graded

Repeat for Credit: N

Multiple Enroll: N

Course Attributes: MSc Physics
MSc Advanced Materials Science

Topics:

COURSE OUTCOMES

Learning Outcomes:

- (1) To learn the basic principles of surface science.
- (2) To know the technologies that are related to thin films.
- (3) To understand the basic principles of vacuum science and technology that are related to surface science and thin-film deposition techniques.
- (4) To understand the thermodynamics of evaporation and learn the basic principles of evaporation.
- (5) To acquire the basic knowledge about adsorption, surface diffusion, nucleation, and structure development that are involved in thin-film deposition.
- (6) To acquire the basic knowledge of epitaxial deposition.
- (7) To learn the basic knowledge of chemical vapor deposition.
- (8) To learn the basic techniques of measuring the physical and chemical properties of thin films.

Course Syllabus:

- 1: Introduction to Surface Science and Thin Film Technology
- 2: Vacuum Science and Technology
- 3: Surface Science
- 4: Evaporation
- 5: Deposition
- 6: Epitaxy
- 7: Chemical Vapor Deposition
- 8: Analysis and Properties of Thin Films

Assessment Type:

Homework or assignment	: 15%
Presentation	: 30%
Test or quiz	: 55%

Feedback for Evaluation:

1. Send email to the teacher, make appointment with the teacher for discussion, or come to his/her office, especially for issues that require immediate action;
2. Express your views in the mid-term and term-end course evaluation;
3. Express your views (or ask a student representative to help you convey your messages) in the staff-student consultation meeting held every year;
4. Post a message on the forum of the course website or on the Discussion Board of the CUHK Blackboard.

Required Readings:

1. John B. Hudson, "Surface Science: An Introduction", Boston: Butterworth-Heinemann, 1992, QC173.4.S94 H84.
2. Kenjiro Oura, Victor G. Lifshits, Alexander A. Saranin, Andrey V. Zotov, Mitsuhiro Katayama, "Surface Science – An Introduction", Springer-Verlag, QC173.4.S94 S96425 2003.
3. Donald L. Smith, "Thin-Film Deposition: Principles & Practice", McGraw-Hill, Inc., TA418.9.T45S65 1995.
4. Milton Ohring, "The Materials Science of Thin Films", Academic Press, TA418.9.T45 047 1992.
5. Anthony C. Jones, Michael L. Hitchman, Ed., "Chemical Vapor Deposition: Precursors, Processes and Applications", RSC, TS695.C67 2009.
6. Hans Luth, "Solid Surfaces, Interfaces and Thin Films", 6th Edition, Springer, QC173.4 S94 L86 2015.
7. Terry L. Alford, Leonard C. Feldman, James W. Mayer, "Fundamentals of Nanoscale Film Analysis", Springer, QC176.83 A44 2007.

Recommended Readings:

OFFERINGS

1. AMSC5738 Acad Organization=MSEGV; Acad Career=TPG

COMPONENTS

LEC : Size=30; Final Exam=Y; Contact=3

ENROLMENT REQUIREMENTS

1. AMSC5738

New Enrollment Requirement(s):

Exclusion = Not for students who have taken PHYS5620.

Other Requirement = For students in MSc Advanced Materials Science

For students in MSc Physics

Additional Information

VTL-Onsite face-to-face hrs 0

VTL-Online synch. hrs 0

VTL-Online asynch. hrs 0

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